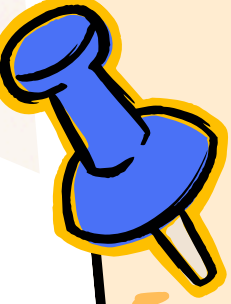


TYPESCRIPT



WHY TYPES MATTER



THE MYSTERY BOX GAME

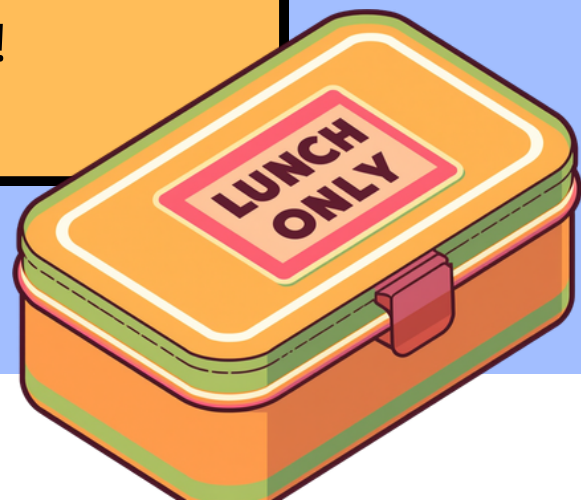


Imagine a Mystery Box That Can Hold Anything!

Think about a magical box with no rules — you could put toys, books, snacks, or even cricket balls inside! But that gets confusing, right?

What If We Want to Keep Only Lunch in It?

What if we put a label on our box saying 'Lunch Only'? Now everyone knows exactly what belongs inside — no more mix-ups!



THE MARKET MYSTERY



JavaScript is Like a Busy Local Market!

Imagine a crowded bazaar with no labels on anything. You ask for sugar, but get salt! That's JavaScript — no labels means surprises!

Sometimes We Buy the Wrong Thing!

Without proper labels, we might mix up numbers with words, or put the wrong things together. Mistakes happen at the last moment!



THE PROBLEM: $10 + '10' = '1010'?$



Why Did Numbers Become Words?

In JavaScript, when you write `10 + "10"`, it joins them like words instead of adding them as numbers. The result is `"1010"` — not 20!

JavaScript Gets Confused Without Labels!

It's like mixing up your math notebook with your Hindi notebook — everything gets jumbled! Without proper labels, JavaScript doesn't know what you really want.



HOW CAN WE FIX THIS? MEET TYPESCRIPT!

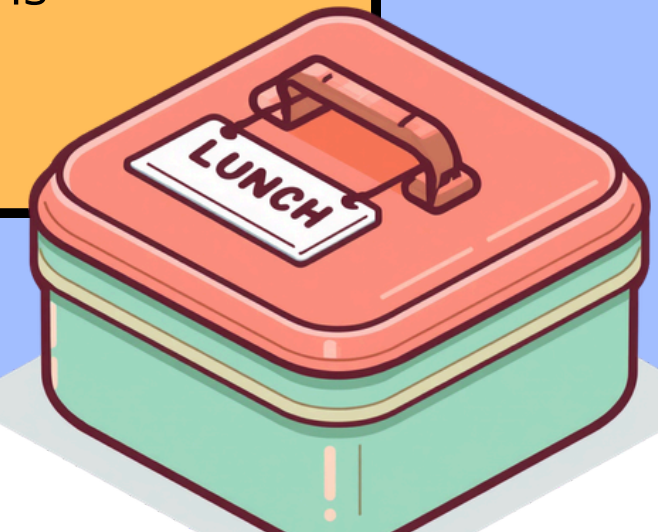


TypeScript Adds Labels to Our Boxes!

Just like labeling your tiffin box helps you know what's inside, TypeScript adds labels to your code. No more surprises or mix-ups!

It Helps Us Catch Mistakes Before Cooking!

Before you serve your code, TypeScript checks if everything is correct. It's like a helpful friend who spots mistakes early!



STATIC VS. DYNAMIC TYPING



Dynamic Typing – The Crowded Market

JavaScript is like a busy local market with no labels. You might pick up sugar thinking it's salt — errors appear only at runtime!

Static Typing – The Organized Store

TypeScript is like Reliance Fresh with clear labels on every shelf. You know exactly what you're getting — errors caught at compile time!



THE LABELED TIFFIN BOX



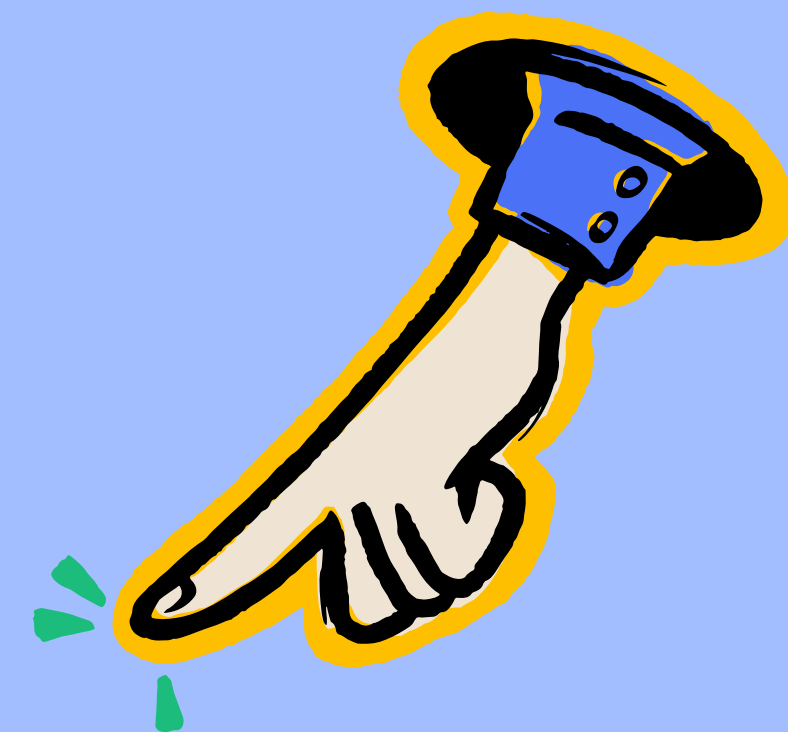
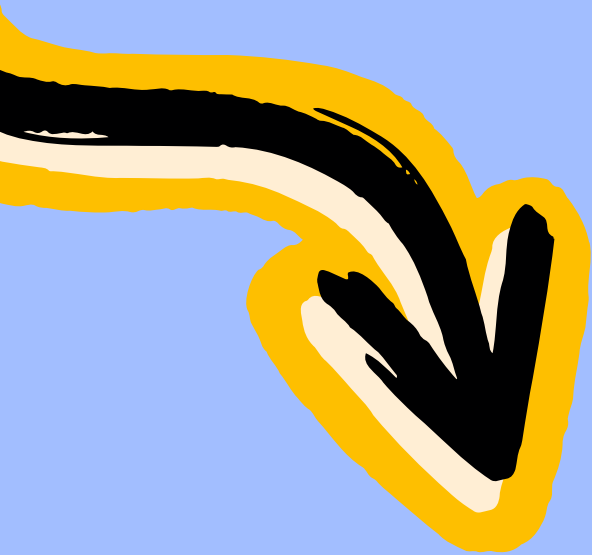
If Tiffin Says 'Lunch Only', No Cricket Ball Allowed!

Just like your tiffin box has a label saying what goes inside, TypeScript uses types to make sure only the right kind of data goes in the right place.

TypeScript Flags Errors If We Try!

When you try to put something wrong, TypeScript shows a red warning before your code even runs. No more surprises!





TYPES YOU SHOULD KNOW

string: City Name like Delhi or Mumbai

number: Roll Number or Marks in exams

boolean: IsHoliday – true or false

Types are labels that tell TypeScript what kind of data to expect!

Let's explore each type!

Typescript TYPES



String Type



Neport Type



Number Type



String Type



Boolean Type

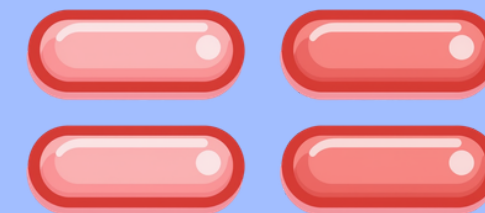


Boolehan Type





HOW TO DECLARE TYPES IN TYPESCRIPT



let city: string

let city: string = "Delhi";
Stores text like city names!

let rollNumber: number

let rollNumber: number = 23;
Stores numbers like your roll number!

let isHoliday: boolean

let isHoliday: boolean = false;
Stores true or false values!



MEET THE TRANSPILER – OUR TRANSLATOR FRIEND



Browsers Only Understand JavaScript

Just like how a translator helps people speaking different languages understand each other, the transpiler converts TypeScript code into JavaScript that browsers can read!

TypeScript Translates Our Labels into JavaScript

Think of it as translating from Hindi to English — the meaning stays the same, but now everyone can understand it. Our type labels help us write better code!

```
import code = Sropt;
type sp = IneJawSrc;
code {
  alipter = typeScri
}
tyngat: nirript
```




INTERACTIVE ACTIVITY – TYPE DETECTIVE!

Aadhaar Number

Is this a string or number? Think about the 12 digits!

Bus Route Name

What type is "Route 42A"? String, number, or boolean?

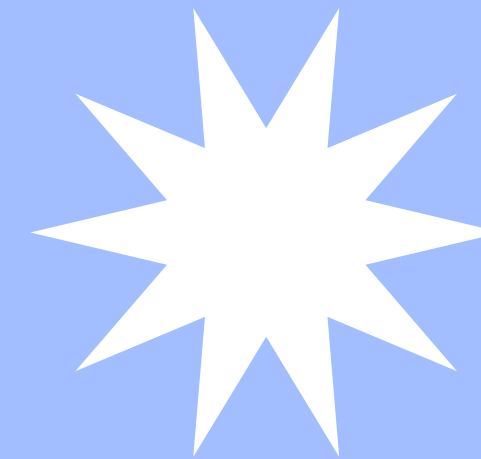
Name 'Zara'

Is 'Zara' a boolean or string? What do you think?





WHY DO TYPES MATTER? RECAP



Catch Mistakes Early

Types help us find errors before our program runs, just like checking your tiffin before school!

Keep Code Clean & Safe

With types, our code stays organized and works exactly as expected every time.

Labels for Our Code

Types are like labels on tiffin boxes — they tell us exactly what belongs inside!



SUCCESS CHECK – WHAT DID WE LEARN?



What happens if we add `5 + '5'` in JavaScript?

Think back to our lesson — does JavaScript combine them as numbers or join them as text? Write your answer!

Name one TypeScript type you learned today!

Remember: string, number, boolean — which one can you explain to a friend? Share your favorite!



Planrs Academy

THANK YOU!

LET'S KEEP EXPLORING!

